Artificial Intelligence

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Print

```
[3]
✓ 0s
print("My name is Umar")
print("I am learning Python")

My name is Umar
I am learning Python
```

Variables

```
[5]
✓ 0s

name = "Umar"
age = 25
city = "Islamabad"

print(name)
print(age)
print(city)

Umar
25
Islamabad
```

Multiple Variables

```
[6]
✓ 0s

x = 5
y = 10
z = x + y

print("Total is:", z)

Total is: 15
```

Data Types

```
[8]
✓ 0s
name = "Umar Farooq"   # string
age = 25        # integer
height = 5.9     # float
isStudent = True # boolean

print(name)
print(age)
print(height)
print(isStudent)

Umar Farooq
25
5.9
True
```

User Input

```
[10]
✓ 7s
name = input("Enter your name: ")
print("Welcome", name)

Enter your name: UMAR FArooq
Welcome UMAR FArooq
```

User Input Numbers

[11]

✓ 3s

```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

sum = num1 + num2

print("Total =", sum)
```

```
Enter first number: 2
Enter second number: 3
Total = 5
```

User Input Numbers Arithmetic Operations

[12]
7s

```python
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

sum = num1 + num2
mul = num1 * num2
sub = num1 - num2
div = num1 * num2

print("Addition =", sum)
print("multiplication =", mul)
print("Subtraction =", sub)
print("Division =", div)
```

```
Enter first number: 3
Enter second number: 1
Addition = 4
multiplication = 3
Subtraction = 2
Division = 3
```

If statement

```
[13]
✓ 0s
age = 18

if age >= 18:
    print("You can vote")

You can vote
```

If Else

```
[14]
✓ 0s
age = 16

if age >= 18:
    print("You can vote")
else:
    print("You cannot vote")

You cannot vote
```

While Loop

```
[15]
✓ 0s
i = 1

while i <= 5:
    print(i)
    i = i + 1

1
2
3
4
5
```

For Loop

```
[16]
✓ 0s
for i in range(1,6):
    print(i)

1
2
3
4
5
```

Functions

```
def greet():
    print("Hello Umar")

greet()
```

```
Hello Umar
```

Functins with parameters

```
[19]
0s
def greet(name):
    print("Hello", name)

greet("Ibrar")
greet("Umar")
```

```
Hello Ibrar
Hello Umar
```

Arrays

```
[20]
✓ 0s
fruits = ["Apple", "Banana", "Mango"]

print(fruits)
print(fruits[0])
print(fruits[1])

['Apple', 'Banana', 'Mango']
Apple
Banana
```

Array using Loop

```
[21]
✓ 0s
fruits = ["Apple", "Banana", "Mango"]

for fruit in fruits:
    print(fruit)

Apple
Banana
Mango
```